

FAIRMODE WG5:

Phase 2 of the “bias projection exercise” - real world data use case

Stijn Janssen, Joana Soares, Anthony Rey-Pommier, Enrico Pisoni, Alexander de Meij, Philippe Thunis, Kees Cuvelier

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1. Context

In this document we propose a follow-up of the previous “WG5 bias projection” exercise, organised in 2025. When developing an air quality plan or roadmap, the new AAQD requires an assessment of absolute concentration levels that can be compared against limit or target values. These concentration levels must be evaluated in an unknown future under specific policy scenarios.

Modelling results are subject to biases, which must be minimised when estimating absolute concentration levels. Such biases can be quantified using observations taken at measurement station, but their generalisation over a modelling domain remains challenging. An additional challenge emerges in model simulations for future scenarios, for which the bias observed in a reference situation must be generalised and “projected” into the future to correct the results of the scenario simulation. This process of bias projection is depicted in Figure 1 below.

Bias projection

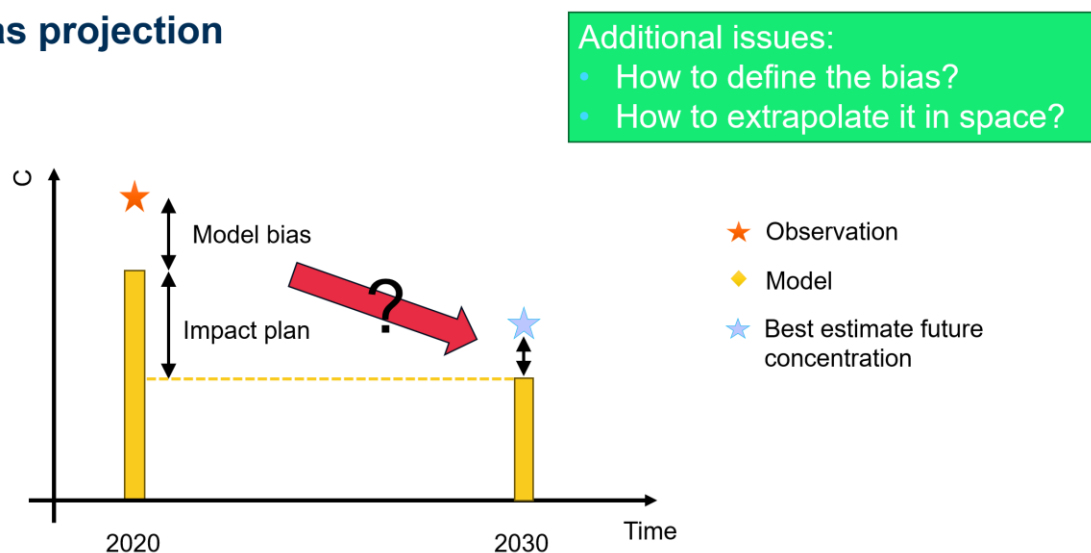


Figure 1: Schematic representation of a bias projection into a future scenario.

Discussions within WG5 made clear that Member States have developed different methodologies to deal with these challenges.

To benchmark these methodologies, a first bias projection exercise, involving reference and projected concentrations derived from a synthetic data set, was performed in early 2025. The

synthetic data was derived from a set of EMEP simulations. To further underpin the findings of the synthetic Phase 1 exercise, WG5 is setting up a second intercomparison exercise for 2026 based on real-world data. For this Phase 2, an intercomparison based on real observations is proposed, providing a known and real-world “ground truth” (observations) for benchmarking.

2. Setup of the benchmarking study

The layout of the Phase 2 in the bias projection benchmarking study is schematically depicted in Figure 2. EMEP model output for the year 2015 is available on the full European domain (available at a yearly/daily resolution). In addition, EMEP simulations for years 2022, 2023 and 2024 are provided as “future scenarios” compared to the 2015 situation. Every year of simulation is obtained using the meteorology and emissions of the corresponding year. In addition to these four simulations, daily observations from EEA monitoring stations in Europe for the same years (2015, 2022, 2023, 2024) are provided to the participants.

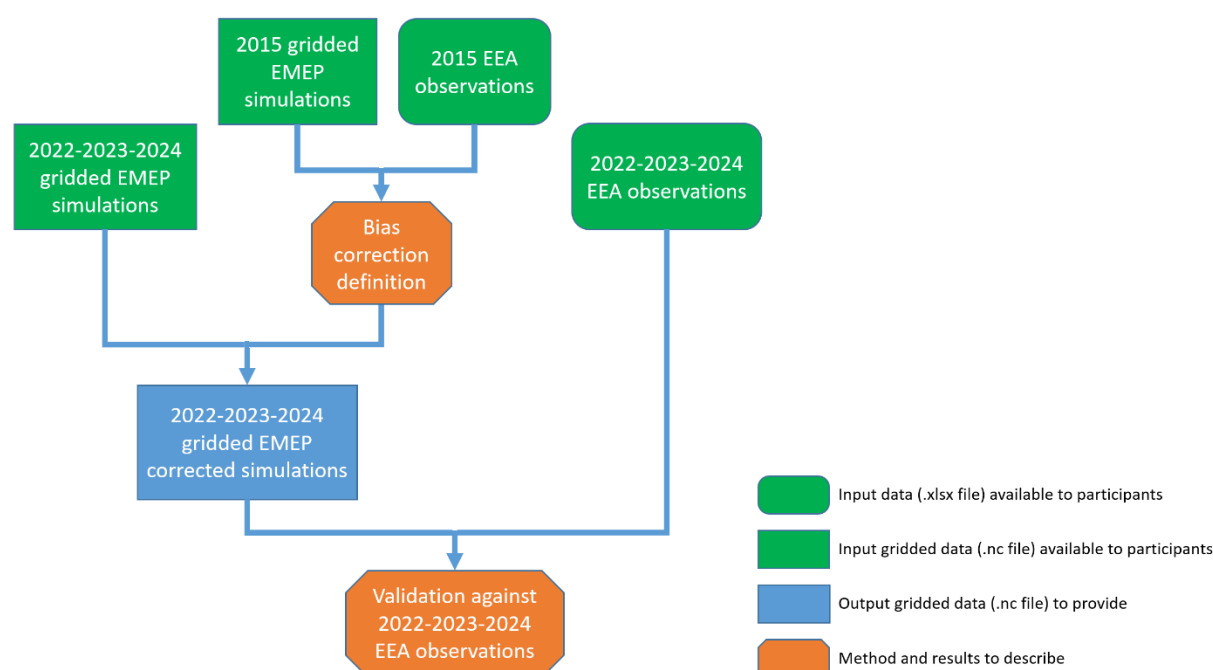


Figure 2: Layout of Phase 2 of the bias projection benchmark study. Input data (2015 gridded EMEP simulations and observations provided by EEA) are displayed in green. The “training phase” (definition of the “bias correction”) allows to define a bias correction, which is then used to generate a gridded file of unbiased concentrations from EMEP simulations for the years 2022, 2023 and 2024. These EMEP concentrations are then compared with EEA observations for the same years during the “validation phase”. The bias projection exercise consists in describing the unbiasing method (orange frames) and delivering unbiased fields (blue frame).

This exercise involves two steps:

- The first step is to define a bias correction for the year 2015, based on EMEP simulations and EEA stations for that year (both daily or yearly values can be used for this purpose);
- The second step is to “project” the bias correction to the EMEP simulations for 2022, 2023 and 2024. This will allow to compare the original (unbiased) EMEP vs the “corrected” EMEP and to assess if the correction nudges the EMEP values closer to

the real-world observations at the EEA stations. This comparison can be done for the 3 available years (2022, 2023, 2024), in order to reflect how meteorological variability impacts bias correction.

The objective of this Phase 2 exercise is to check if the recommended “best approach” for bias projection (as derived by the synthetic dataset in Phase 1) is confirmed when using a “real-world” dataset.

Modelling results will be available for PM_{2.5}, NO₂ and O₃ as daily values for the full year and as an annual means.

3. Description of the exercise

To participate in the benchmarking exercise, the following steps must be performed:

1. Download all required datasets from the JRC server (EMEP model results, EEA observations):
2. Derive a “bias correction” for the year 2015, based on your preferred methodology, using daily or yearly values from EMEP model fields and EEA observations. Note that in this procedure, no new model simulations must be performed. Only post-processing of the available modelling results is required.
3. Apply your preferred bias projection methodology to the years (2022–2024) to obtain bias-corrected EMEP results (under changing meteorology and emissions). No new model simulations must be performed: only post-processing of the provided fields is required.
4. Provide to JRC the gridded NetCDF files computed in the previous step, containing annual means of PM_{2.5}, NO₂ and O₃ for the entire European domain, for both the reference year (2015) and projected years (2022–2024).
5. Briefly describe the bias correction and bias projection methodology applied, for further reference and interpretation in the exercise.

4. Description of the data sets

The data is available at a JRC server via this [Teams group](#). Note that a login via a known email address is required. Under the “File > DATA input for participants” folder, two data sets are available: one for daily values, one for yearly values. Participants are free to work with the daily or the annual mean values. Each time aggregation contains the following structure:

- BaseCase_2015_Gr idded
- BaseCase_2015_Points
- Scenar io_2022_Gr idded
- Scenar io_2023_Gr idded
- Scenar io_2024_Gr idded

Within each folder, data is available for the three pollutants PM_{2.5}, NO₂ and O₃. “Gridded” data is provided as .nc files, “point” data as .csv files.

5. Data delivery

The data delivered under Step 5 should be annual averages for the entire European domain. The structure of the NetCDF file should be as in the original yearly simulated fields provided by the JRC (2D array).

The data can be uploaded at the JRC server via the dedicated [Teams group](#) (under the "File > DATA contributions from participants" directory) where an institute folder can be created. In case of access problems contact WG5 leads and/or JRC.

To allow for semi-automated data processing, file naming (of the uploaded files) should follow the structure:

- Scen_<year>_<group>_<species>_<info>_CORR_YEARLY (extension .nc or .tiff)

with:

- <year>: projection year 2022, 2023 or 2024
- <group>: acronym of the modelling team
- <species>: NO2, O3, PM25
- <info>: info on the bias projection method (see further)

The <info>-tag can contain information regarding the applied bias projection methodology, all subitems separated by a dot. To this extent, the framework developed by the Bonafè et al. (see this [link](#)) is very instructive and might help to further classify the methodologies. Participants are requested to map their methodology onto the generic structure:

<sequence>.<calibration>.<correction>.<spatialization>

with

- <sequence> is one of SCA, CSA, CAS, CA
- <calibration> is one of Each, All, Cell, Grid, Neigh
- <correction> is one of Add, Mult, Lin
- <spatialization> is empty (for CA sequence) or one of TPS, IDW, OK, KED...

For details, see the link above.

In case of doubt, another description can be used in the <info>-tag. A short document describing addition meta information of the bias projection methodology can be uploaded along the data sets into the Institute folder.

6. Timing

The following timeframe is proposed:

- January 15, 2026: expression of interest to participate
- End of January 2026: the data is ready for download
- February - April 2026: the exercise is undertaken
- March 2026: Presentation of first results at the Budapest FAIRMODE Plenary Meeting